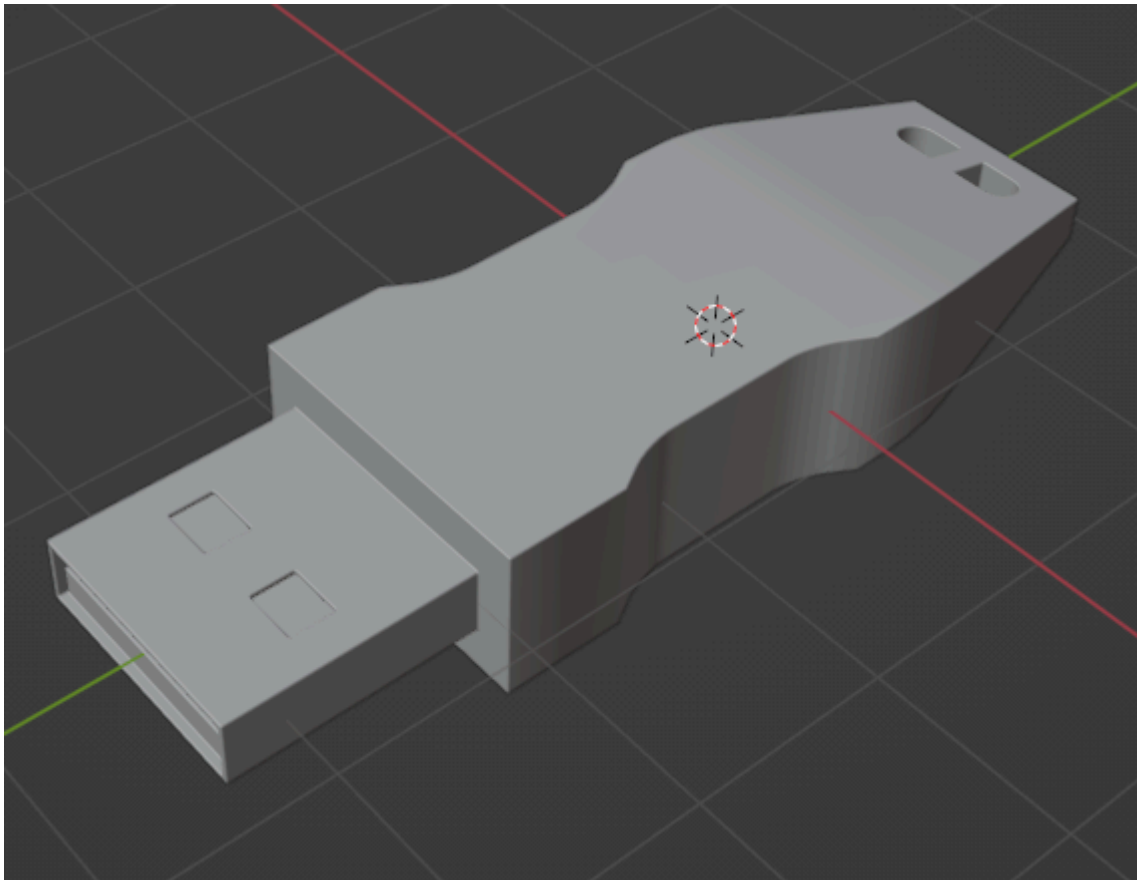
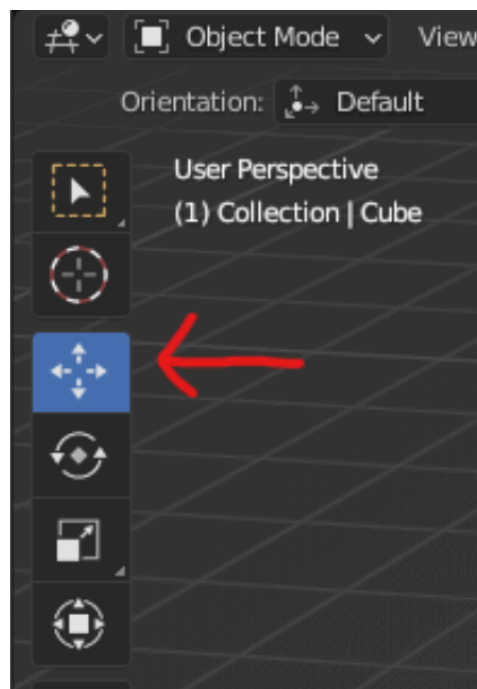


Difficult tutorial 2

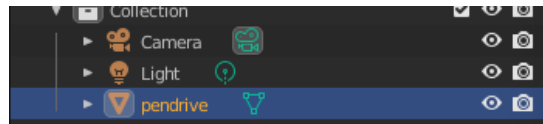


In this exercise, we are going to create a pendrive.

To start, set the viewport mode to show arrows of the axis to aid in the modeling process:

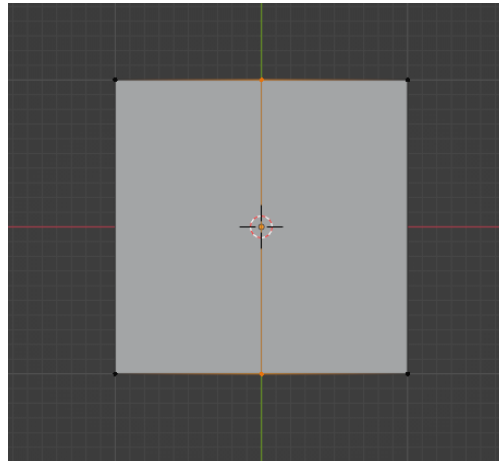


Then, rename the cube to “pendrive”:

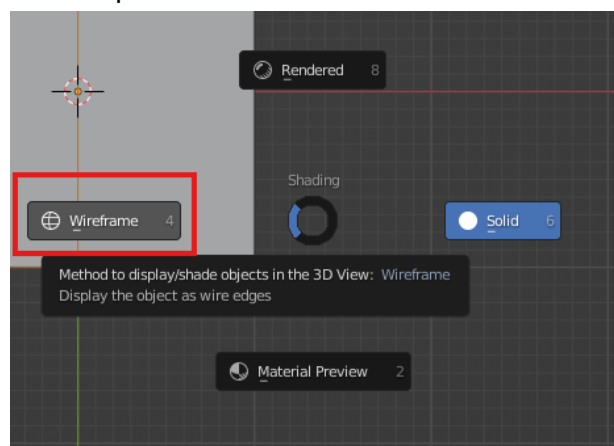


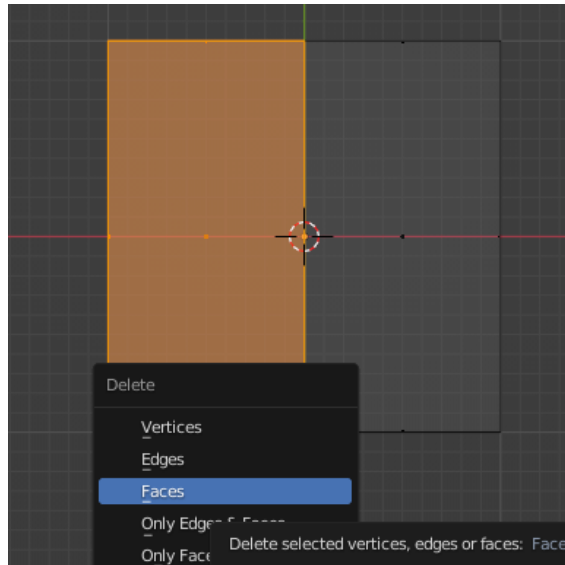
Enter edit mode (TAB) and go to the top view.

Add a loop cut (CTRL+R) in the middle of the cube:

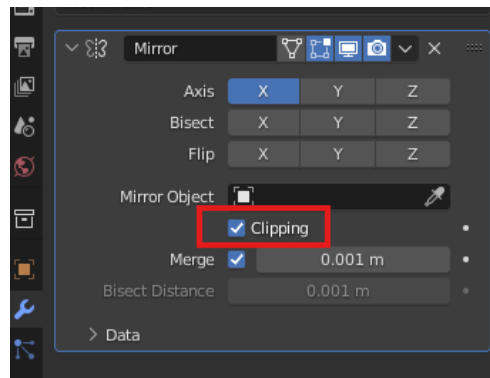


Go to wireframe shade mode (press Z and select Wireframe), change to face selection mode (press 3) and select all the left part of the cube and delete it:

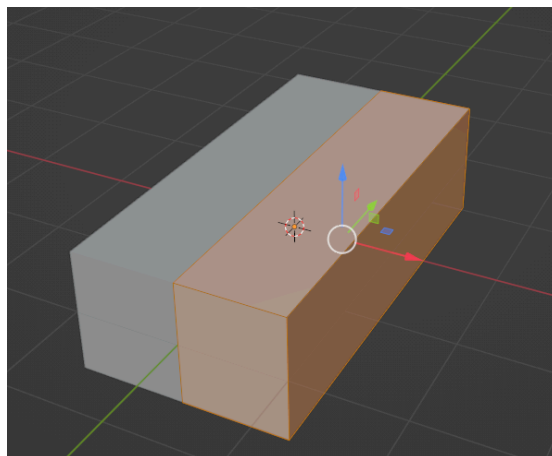




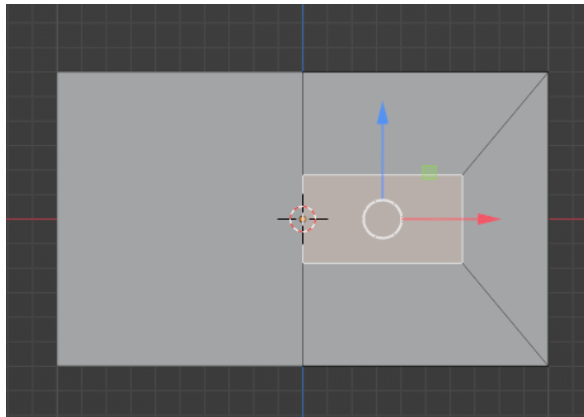
With the left part deleted, go to the modifiers panel and add a mirror modifier, enabling also Clipping:



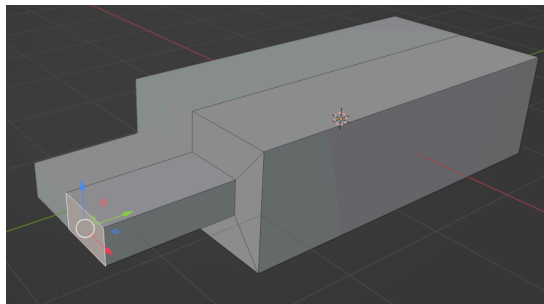
Now go back to solid shading mode, select all and scale along Z axis (S+Z) by 0.6 and along Y axis (S+Y) by 2:



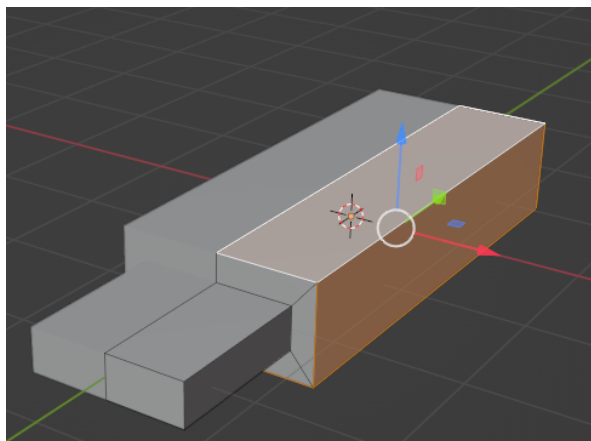
Go to the front view (NUMPAD 1), select the face, extrude it (E+ENTER) and scale (S) by 0.3:



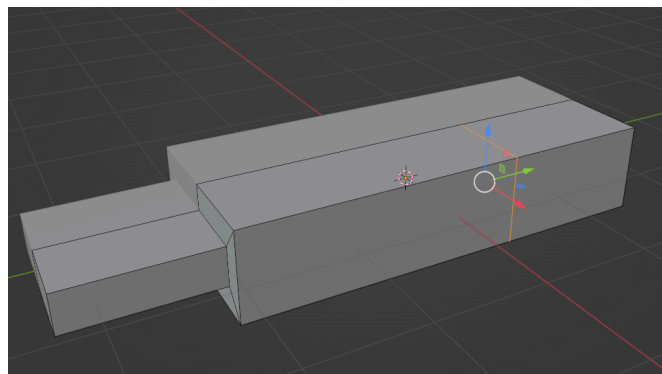
Now extrude again (E+ENTER) by -1.3 in the Y direction (E+Y):



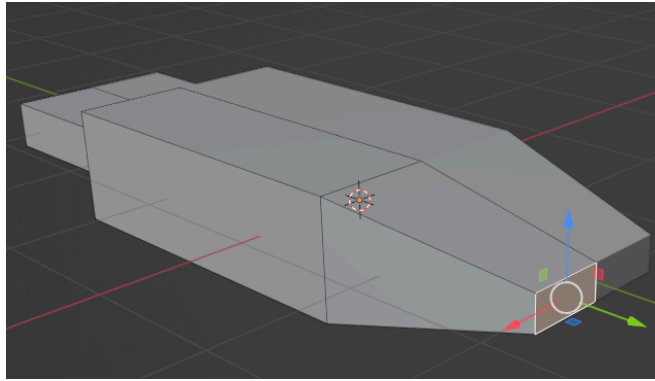
Now, select the outer faces of the body of the pendrive and scale it in all directions **except** the Y (S+SHIFT+Y) by 0.7:



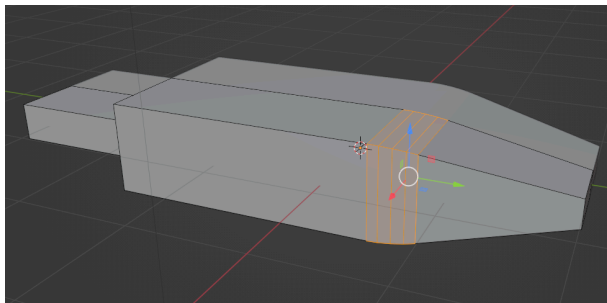
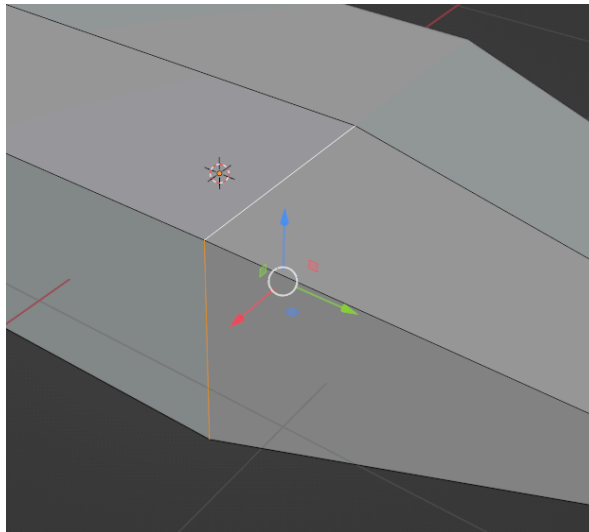
Now add a loop cut (CTRL+R) on the body of the pendrive setting 0.3 as a factor:



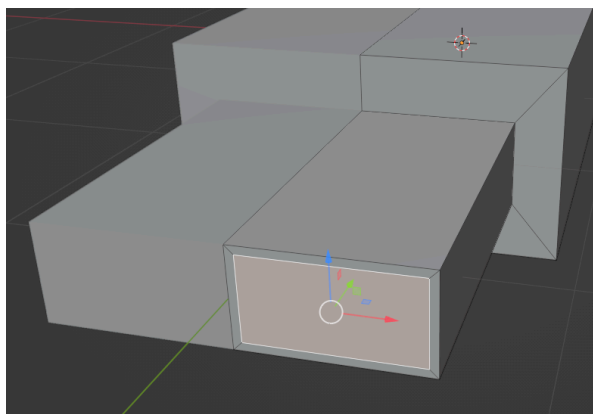
Select the bottom face of the body and scale it (S) by 0.3:



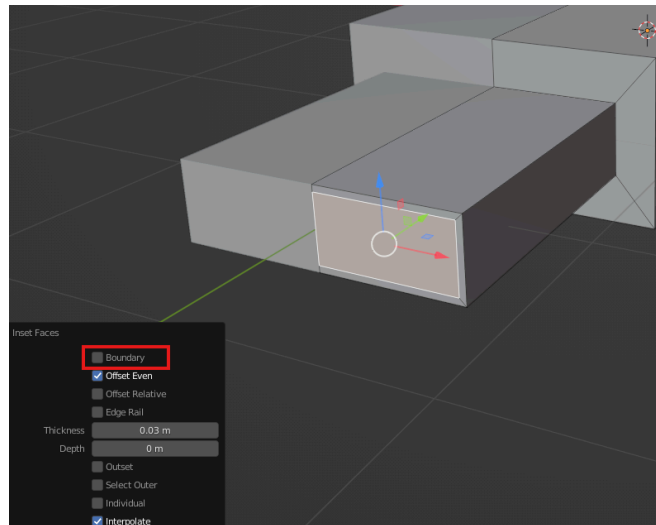
Now go to edge select mode (press 2) and select the edges created in the middle of the body previously by loop cut and bevel it (CTRL+B) by 0.2 and 4 segments:



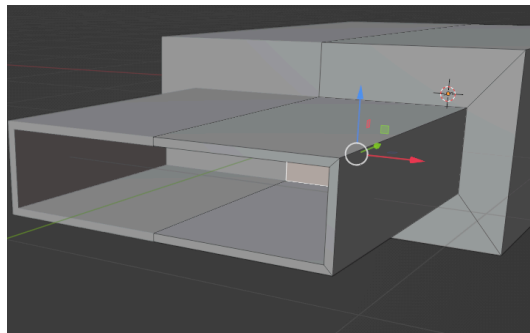
Now select the top face of the USB port and inset it (I) by 0.03:



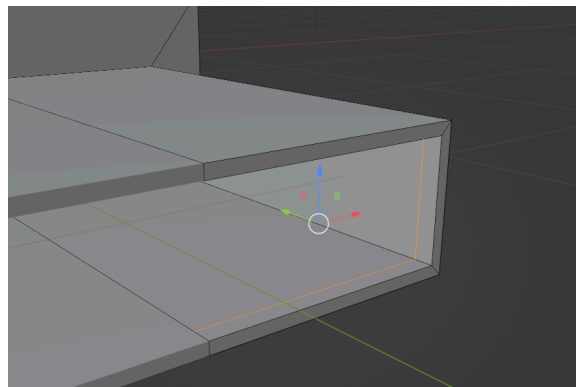
Open the small dialog in the bottom left corner of the screen and deselect “Boundary” under options:



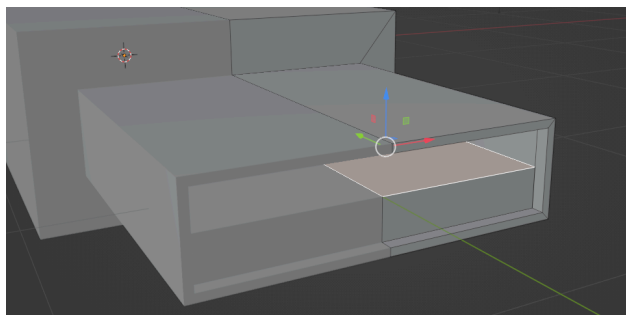
Now extrude it (E+ENTER) and move on Y axis (G+Y) by 1:



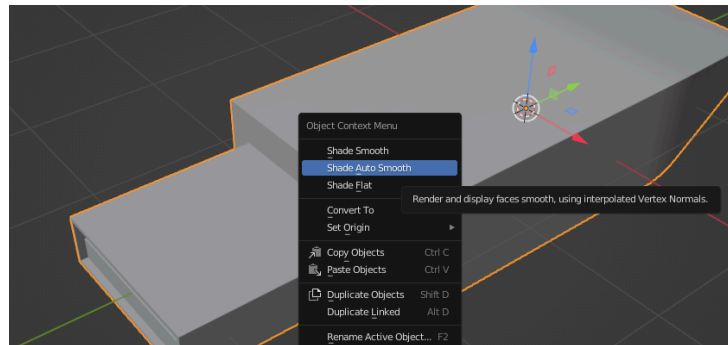
Now add a loop cut (CTRL+R) inside the USB port offsetting it by 0.9:



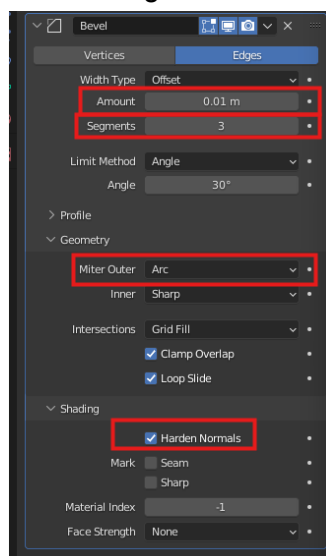
Now go to face selection mode and select the bottom face inside the USB port and extrude it (E+ENTER) on the Z axis (G+Z) by 0.15:



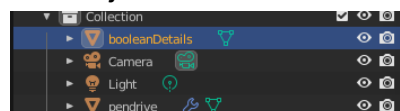
Now go to object mode (TAB), with the pendrive object selected, right click the viewport and select “Shade Auto Smooth”:



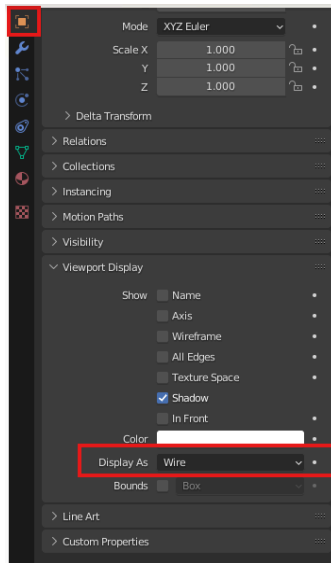
Add a Bevel modifier and set the Amount to 0.01, segments to 3, under Geometry set the Miter Outer to “Arc” and finally under Shading set “Harden Normals” to True:



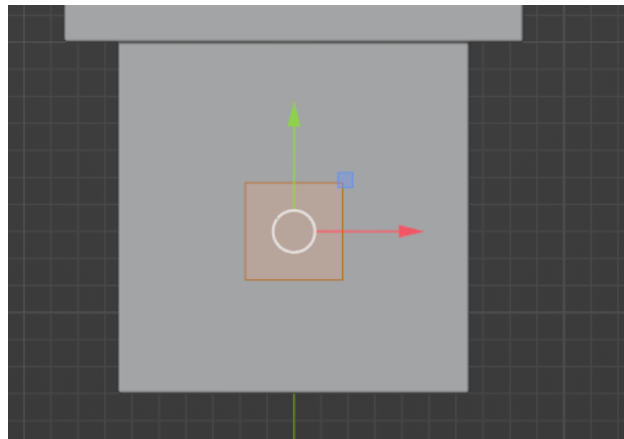
Now, make sure that the 3D cursor is at the center of the scene and add a cube (SHIFT + A → MESH → Cube) and rename the object to “booleanDetails”:



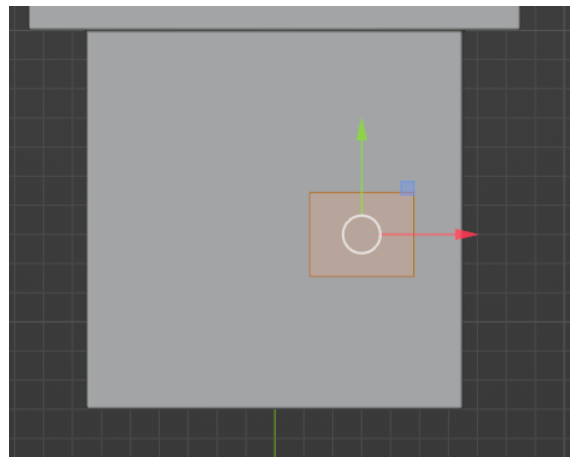
Under Object properties > Viewport Display, set “Display As” to “Wire”:



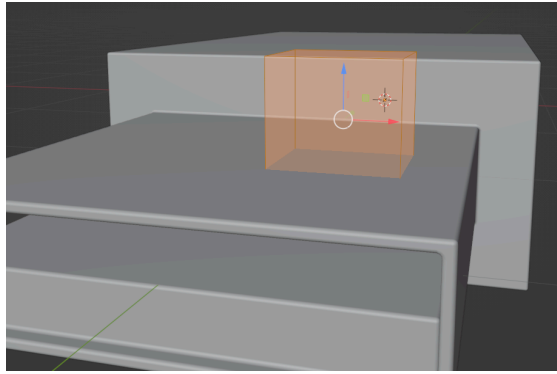
Enter edit mode (TAB), move the cube along Y axis (G+Y) by -2.7, scale the cube (S) by 0.18:



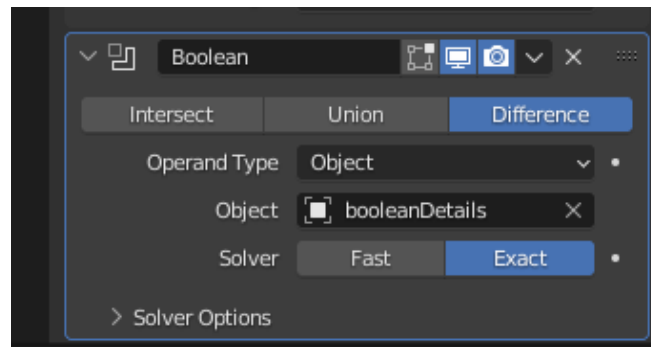
Then, scale on the Y axis (S+Y) by 0.8 and move the cube on the X (G+X) axis by 0.3:



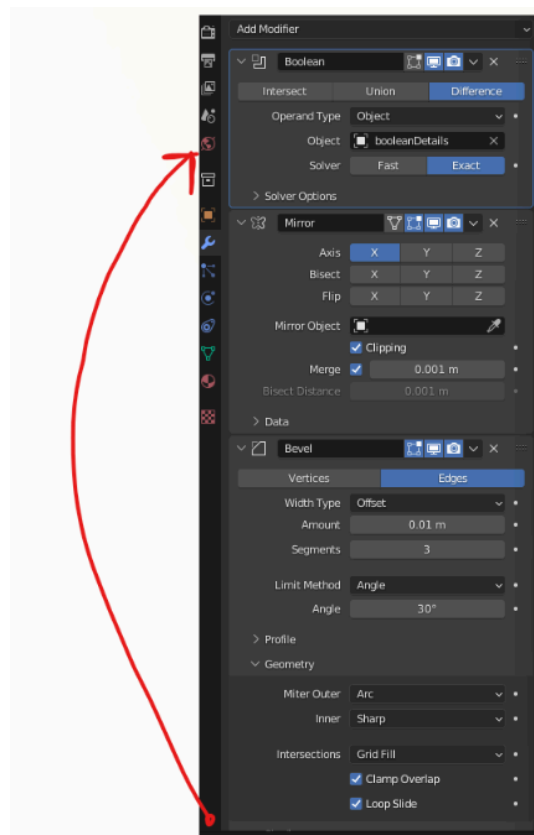
Now move it in the Z axis (G+Z) by 0.3, so it is just passing through the outline of the USB port:



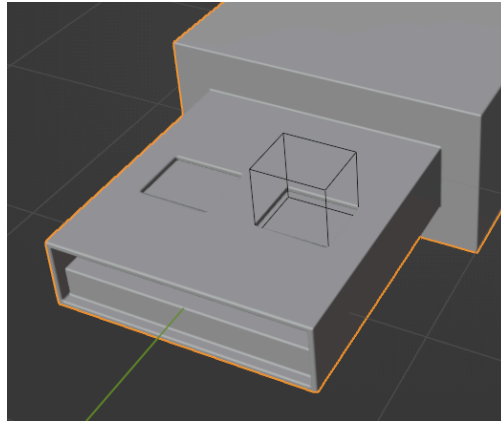
Now change to object mode (TAB), select the pendrive object and add a Boolean modifier, selecting the “booleanDetails” object as the target object:



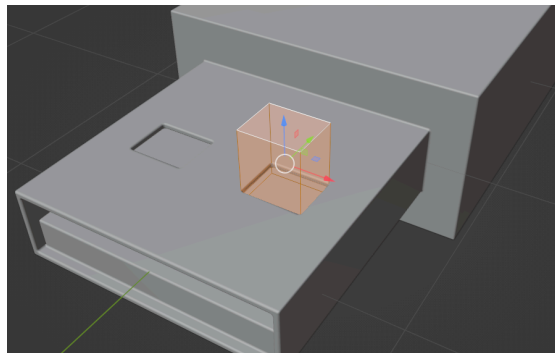
Next, move the Boolean modifier up, so it is the first modifier applied:



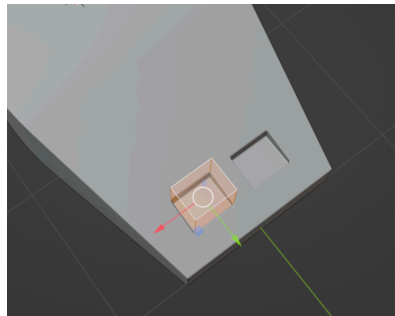
Now as you can see, the details object will cut through the mesh and it will be mirrored:



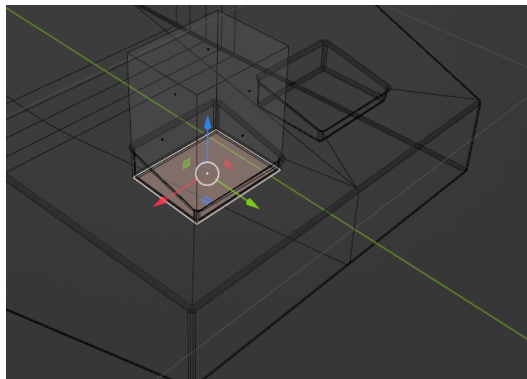
Select the “booleanDetails” object and enter edit mode (TAB).
Select the cube and scale it in all directions except the Z direction (S+SHIFT+Z) by 0.8, so the hole is a little bit smaller:



Now, let's add a hole in the back of the pendrive body so a rope can be attached:
Copy the cube (SHIFT+D) and move it in the Y axis (G+Y) by 4.4. Then, move it in the X axis (G+X) by -0.08:

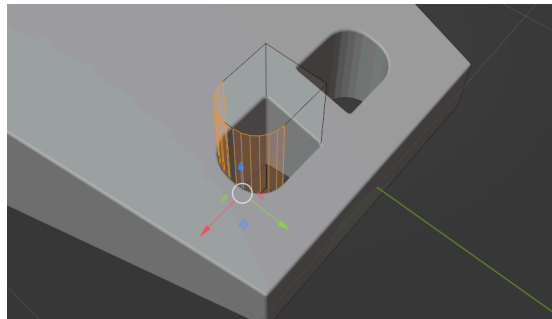
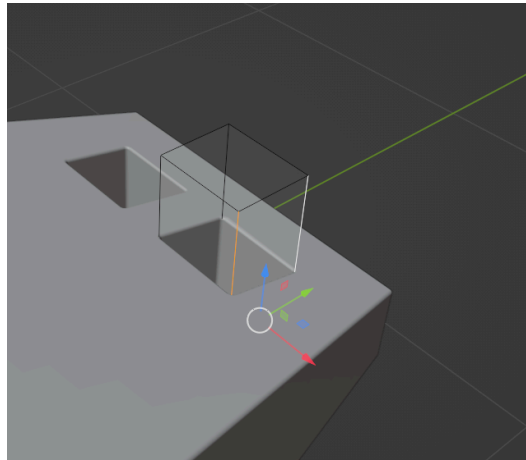


Now, to make it cut through, we have to move its bottom face down. To make this, change to wireframe mode (press Z and select Wireframe) and select the bottom face:



Move it along the Z axis (G+Z) by -0.5 (so it cuts through the whole body) and go back to Solid shading mode (press Z and select Solid).

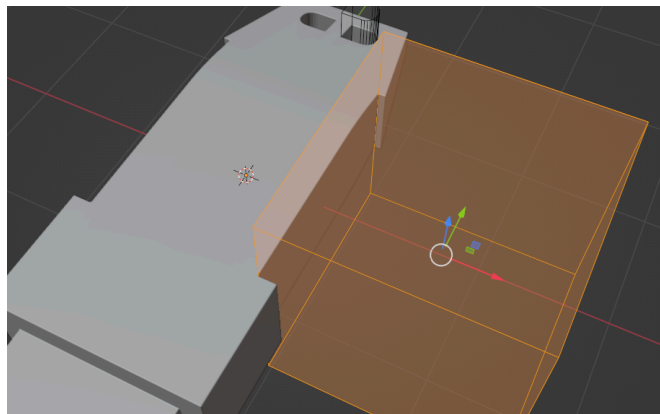
Select outer vertical edges and bevel it (CTRL+B) by 0.1 and 6 segments, so we have this:



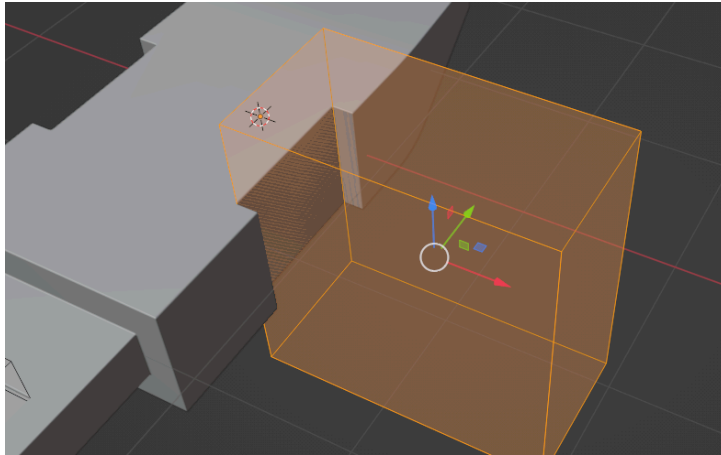
To add a final touch, let's create an ergonomic detail on the body of the pendrive:

Make sure that the 3D cursor is at the center of the scene.

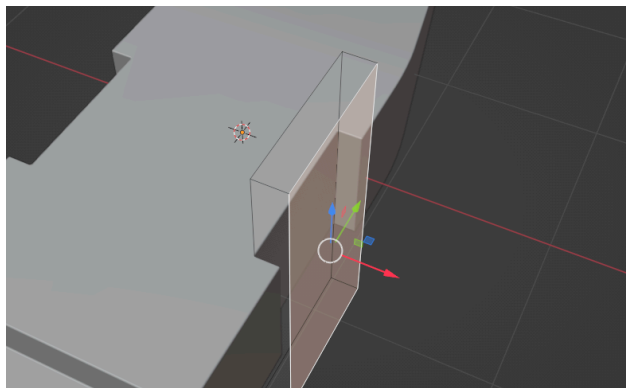
Still in the edit mode, add another cube (SHIFT+A → Mesh → Cube) and move it along the X axis (G+X) by 1.7:



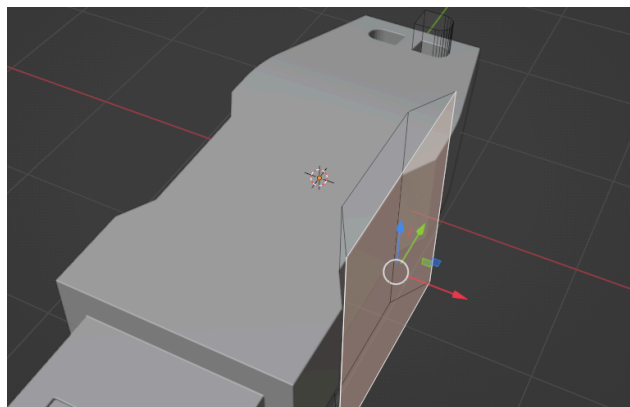
Scale it in the Y axis (S+Y) by 0.5 and move it along the Y axis (G+Y) by -0.6:



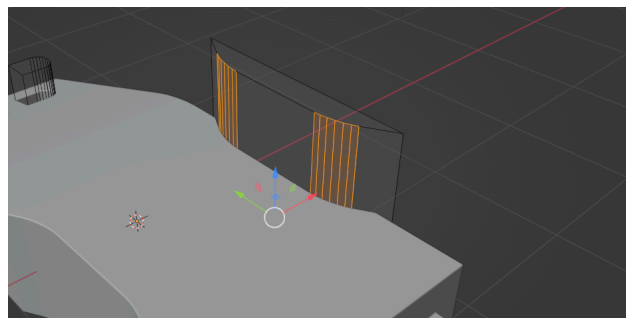
Change to face mode (press 3) and select the farthest face from the pendrive and bring it closer by moving it along the X axis (G+X) by -1.8:



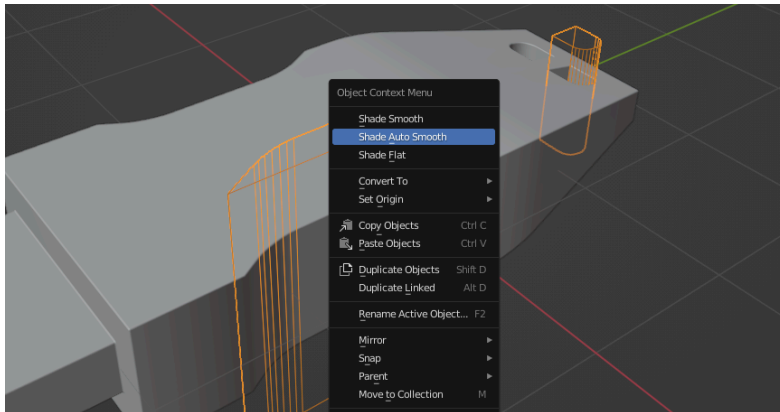
Now scale this face on the Y axis (S+Y) by 1.7:



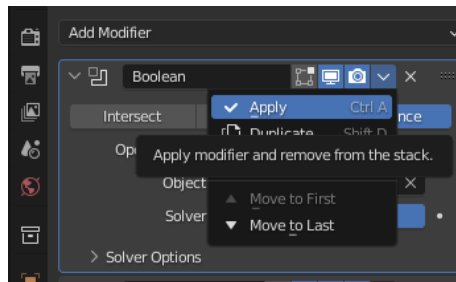
Now select the innermost vertical edges and bevel them (CTRL+B) by 0.15 again with 6 segments:



Now, go to object mode (TAB), right click in the viewport and select “Shade Auto Smooth”:



Finally, select the pendrive and apply the Boolean modifier:



Now you can delete the “booleanDetails” object to clean up the scene.

